Adaptive Per-Distance Sampling for Efficient Multi-Scale Network Centrality Analysis

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Abstract

Computing network centrality at scale is expensive. Sampling can help, but how much is enough? We present a practical model: maintain an effective sample size of $n_{\text{eff}} = \max(k \cdot \sqrt{r}, n_{\min})$, where r is network reach, to achieve ranking accuracy $\rho \geq 0.95$. The proportional term ($k = 9.91$) handles large networks; the floor ($n_{\min} = 350$) ensures reliability at low reach. The model validates on a 294,000-node Greater London street network, achieving $\rho > 0.98$ at distances from 5–20km with speedups of 5.3–21.0×. The implementation is available in the open-source `cityseer` Python library.

Keywords: network centrality, sampling, urban morphology, betweenness, harmonic closeness, computational efficiency

1 Introduction

Network centrality measures are fundamental tools for understanding urban street networks, revealing patterns of accessibility, connectivity, and flow that shape how cities function. Closeness centrality quantifies how effectively a location can reach all others, while betweenness centrality captures the extent to which a location lies on shortest paths between other pairs of nodes. These measures have proven invaluable for urban planning, retail location analysis, and understanding pedestrian movement patterns.

However, computing network centrality at scale presents a significant computational challenge. For large urban networks—which routinely contain hundreds of thousands of nodes—exact computation becomes prohibitively expensive, particularly when analysis must be repeated across multiple distance thresholds or network configurations.

Sampling-based approximation offers a natural solution: rather than computing exact centrality from all source nodes, we compute from a random subset and scale the results accordingly. This approach can dramatically reduce computation time while preserving the essential ranking information that practitioners need. The critical question is: *how much sampling is enough?*

This paper presents a practical, empirically-validated answer to that question. We derive a two-parameter model that determines the required effective sample size n_{eff} as a function of network reach r—the mean number of nodes within the analysis distance. The model combines a proportional scaling term ($n_{\text{eff}} \propto \sqrt{r}$), grounded in sampling theory, with a minimum floor that ensures reliability at low reach. The resulting sampling probability $p = n_{\text{eff}}/r$ decreases as networks grow, delivering the largest speedups precisely where computational cost is highest.

We fit the model on synthetic networks spanning three topologies and validate it on real-world street networks in Greater London (294,000 nodes) and Madrid, achieving Spearman $\rho > 0.98$ at distances from 5–20km with speedups of 5.3–21.0×. Formal model comparison across five candidate functional forms and a bootstrap hypothesis test on the scaling exponent support the proposed specification. The model is implemented in the open-source `cityseer` Python library.

The remainder of this paper is organised as follows. Section 2 reviews related work. Section 3 describes our experimental methodology. Section 4 derives and selects the empirical model. Section 5 presents the adaptive sampling algorithm. Section 6 validates the approach on real-world networks. Section 7 provides practical guidance. Sections 8 and 9 discuss limitations and conclude.

2 Background and Related Work

2.1 Network Centrality in Urban Analysis

Network centrality measures have become essential tools for urban morphology and spatial analysis. Originally developed in social network analysis (Freeman, 1977, 1979), these measures were adapted for urban street networks by Hillier and Hanson (1984) and subsequently formalised in the space syntax tradition.

Closeness centrality measures how efficiently a node can reach all others. The harmonic variant, which we use throughout this paper, is defined as:

$$C_H(v) = \sum_{u \neq v} \frac{1}{d(v,u)} \tag{1}$$

where $d(v,u)$ is the shortest path distance from v to u. The harmonic formulation handles disconnected graphs naturally and is preferred for distance-weighted urban analysis (Porta et al., 2006).

Betweenness centrality measures the extent to which a node lies on shortest paths between other pairs:

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}} \tag{2}$$

where σ_{st} is the number of shortest paths from s to t, and $\sigma_{st}(v)$ is the number passing through v. High betweenness indicates locations that control flow through the network.

For urban applications, these measures are typically computed with a distance threshold $d_{\max}$, considering only nodes within a specified radius. This reflects the reality that pedestrian accessibility is local rather than global. Common thresholds range from 400m (5-minute walk) to 20km (regional connectivity).

2.2 Computational Complexity

The computational cost of centrality measures scales poorly with network size. Brandes' algorithm (Brandes, 2001) computes both betweenness and harmonic closeness in $O(NM + N^2 \log N)$ time for weighted graphs, where N is nodes and M is edges. For a network of 294,000 nodes with average degree 3, this translates to billions of operations per distance threshold.

When analysis requires multiple distance thresholds—as is common in urban morphology studies—the computational burden multiplies. A typical multi-scale analysis with 12 distance thresholds on a large network can require days of computation.

2.3 Sampling Approximation

Sampling-based approximation of centrality has been studied extensively. Brandes and Pich (2007) showed that betweenness can be approximated by computing from a random subset of source nodes and scaling appropriately. Subsequent work established theoretical bounds on approximation quality (Bader et al., 2007; Riondato and Kornaropoulos, 2016).

The key insight is that centrality computed from k random sources produces estimates with standard error proportional to $1/\sqrt{k}$. This suggests that effective sample size—the product of reach and sampling probability—should be the key determinant of accuracy.

More recently, adaptive algorithms have been developed that adjust sampling online. KADABRA (Borassi and Natale, 2019) uses progressive sampling with stopping rules based on confidence intervals, providing tighter approximation guarantees for individual queries. Our approach differs in providing *a priori* sampling probabilities based on network reach, enabling practitioners to predict computational cost before execution. The two approaches are complementary: KADABRA provides tighter bounds for specific queries, while our model enables efficient batch computation across multiple distances.

Despite these advances, prior work has focused on theoretical worst-case bounds or online adaptation rather than empirical calibration for practical urban network analysis. The question of *how many samples are sufficient* for typical urban networks has not been systematically addressed. Our work fills this gap by empirically validating that effective sample size determines accuracy across topologies, fitting practical thresholds for a target accuracy of $\rho \geq 0.95$, and deriving a simple model that analysts can apply directly.

2.4 Localised Centrality

Urban network analysis typically uses *localised* centrality measures with a distance cutoff. For a cutoff $d_{\max}$, we compute:

$$C_H(v; d_{\max}) = \sum_{u:d(v,u) \leq d_{\max}} \frac{1}{d(v,u)} \tag{3}$$

The *reach* r at cutoff $d_{\max}$ is the mean number of nodes within that distance:

$$r = \frac{1}{N} \sum_{v} |\{u : d(v,u) \leq d_{\max}\}| \tag{4}$$

Reach varies dramatically with distance threshold. In our Greater London validation network, reach ranges from 2,783 at 5km to 43,165 at 20km—more than an order of magnitude. This variation is central to our sampling model: larger reach allows more aggressive sampling while maintaining accuracy.

3 Methods

3.1 Experimental Design

Our analysis proceeds in two stages: (1) model fitting on synthetic networks with controlled topology, and (2) validation on large real-world street networks.

We generate synthetic networks using 3 topology templates—trellis (high connectivity, representing planned urban cores), tree (low connectivity, representing suburban sprawl), and linear (intermediate connectivity)—that span the range of urban street network structure. Each topology is tiled to create networks of approximately 12km extent. We test 12 distance thresholds from 250m to 4,000m, and 22 sampling probabilities from 2.5% to 100%.

A critical methodological consideration is the treatment of boundary effects. Nodes near network edges have artificially lower centrality because their catchments are truncated. We address this by applying an inward buffer from the convex hull of all nodes, equal to the maximum analysis distance (4km for synthetic networks, 20km for validation). Only nodes inside this buffered zone are marked as "live" and included in accuracy calculations, ensuring all live nodes have complete catchments at all tested distances. This approach concurs with best practice for localised network analysis (Gil, 2017).

3.2 Accuracy Metric

We measure sampling accuracy using Spearman rank correlation (ρ) between sampled and ground-truth centrality values:

$$\rho = 1 - \frac{6 \sum_i (R_i^{\text{sampled}} - R_i^{\text{true}})^2}{N(N^2 - 1)} \tag{5}$$

where R_i denotes the rank of node i. Spearman correlation is appropriate because urban applications typically care about *relative* rankings rather than absolute values, it is robust to the systematic scaling that sampling introduces, and values near 1.0 indicate preserved rankings even if magnitudes differ. We fit the model using a target of $\rho \geq 0.95$: the target accuracy determines which sampling probabilities are deemed sufficient at each configuration, and thus drives the fitted values of k and $n_{\min}$ (Section 4). A stricter target produces larger k (more conservative sampling); a more relaxed target produces smaller k (more aggressive sampling). The methodology generalises to other tolerances, which would yield different model parameters.

3.3 Model Fitting

For each (topology, distance) combination, we identify the minimum sampling probability p^* that achieves $\rho \geq 0.95$ and compute the implied proportional constant $k_{\text{implied}} = p^* \cdot \sqrt{r}$. This formulation arises from sampling theory: standard error scales as $1/\sqrt{n}$, so maintaining consistent precision as reach grows requires $n_{\text{eff}} \propto \sqrt{r}$, giving $p = k/\sqrt{r}$. We take the 75th percentile of k_{implied} across all configurations, balancing coverage against practical speedup.

The proportional model breaks down at low reach, where the implied n_{eff} may be too few samples for reliable estimation. We determine a minimum floor $n_{\min}$ using logistic regression of success probability on effective sample size, inverting the fitted function to find the n_{eff} where predicted success probability reaches 95%. Full fitting details and parameter uncertainty are presented in Section 4.

3.4 Validation Networks

We validate the fitted model on two real-world street networks. The primary validation uses the Greater London Authority (GLA) road network, sourced from Ordnance Survey Open Roads data. After preprocessing—removing filler nodes, consolidating nearby nodes, and removing dangles—the network comprises approximately 294,000 nodes. Its scale far exceeds our synthetic training data, it contains diverse topology (grid-like central areas, tree-like suburbs, and mixed intermediate zones), and reach values at tested distances (5km, 10km, 20km) extend well beyond synthetic data ranges. We apply a 20km inward buffer to eliminate edge artifacts. As an external validation, we also test the model on the Greater Madrid metropolitan street network following the same protocol.

4 Empirical Accuracy Models

4.1 The Opportunity: Sampling Can Work

Figure 1 illustrates why sampling-based centrality is attractive. Panel A shows that accuracy (Spearman ρ) increases with sampling probability across all distance thresholds, but the curves flatten well before 100% sampling. At 4000m distance (reach ≈ 2000), $\rho \geq 0.95$ is achieved at just 20% sampling—a 5× potential speedup.

Panel B quantifies the speedup potential at each distance threshold when targeting $\rho \geq 0.95$. Speedup increases with distance because larger reach allows more aggressive sampling. At short distances (250m), little speedup is possible; at 4000m, speedups exceed 5×.

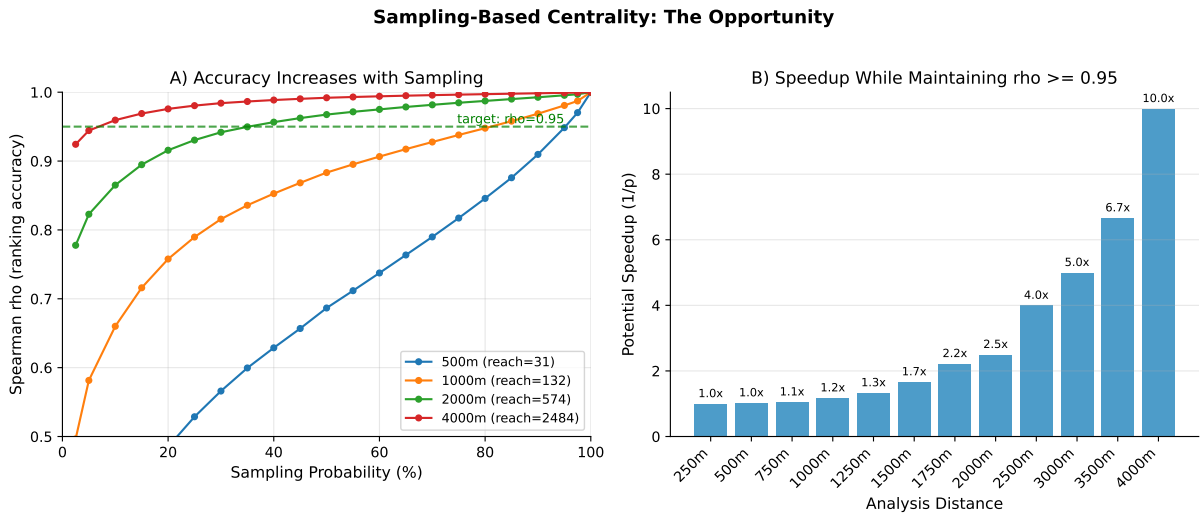

Figure 1: **The sampling opportunity.** (A) Accuracy vs. sampling probability for different analysis distances. The green dashed line marks the $\rho = 0.95$ target. (B) Potential speedup at each distance while maintaining $\rho \geq 0.95$.

4.2 Effective Sample Size: A Unifying Quantity

To derive our sampling model, we first observe that accuracy depends on *effective sample size* $n_{\text{eff}} = r \times p$, where r is the mean reachability (average number of nodes reachable per source within the distance threshold) and p is the sampling probability, not on reach or probability individually. Figure 2A demonstrates this: when we plot all observations (across topologies and distances) as ρ vs. n_{eff}, they collapse onto a single relationship.

This observation simplifies our task: rather than modelling accuracy as a function of both reach and probability, we need only determine the required n_{eff} as a function of reach. The question becomes: how does required n_{eff} scale with network size?

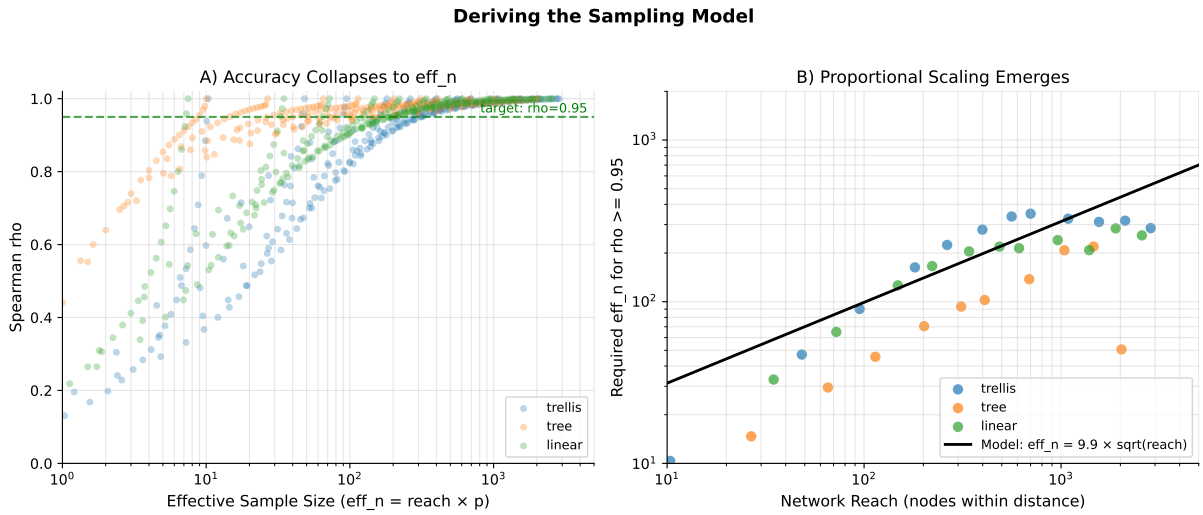

Figure 2: **Deriving the sampling model.** (A) Accuracy collapses to effective sample size: all observations fall on a common curve regardless of topology. (B) Required n_{eff} for $\rho \geq 0.95$ scales as $\sqrt{r}$, motivating the proportional model.

4.3 Proportional Scaling: Why $\sqrt{r}$?

Figure 2B shows the minimum n_{eff} required to achieve $\rho \geq 0.95$ plotted against reach. The relationship is approximately $n_{\text{eff}} \propto \sqrt{r}$, which we parameterise as:

$$n_{\text{eff}} = k \cdot \sqrt{r} \tag{6}$$

This scaling is consistent with sampling theory: maintaining consistent relative precision as reach grows requires sample size proportional to $\sqrt{r}$, since larger networks exhibit more averaging of local variation. To validate this empirically, we fit the general power model $n_{\text{eff}} = k \cdot r^{\alpha}$ via OLS on log-transformed data ($R^2 = 0.73$). The estimated exponent $\hat{\alpha} = 0.5529$ (bootstrap 95% CI: $[0.40, 0.66]$) does not differ significantly from 0.5 ($p = 0.38$), supporting the square root specification (Figure 8).

Fitting k as the 75th percentile of implied k values across all 35 synthetic configurations yields $k = 9.91$.

4.4 The Floor: Reliability at Low Reach

The proportional model $n_{\text{eff}} = k \cdot \sqrt{r}$ breaks down at low reach. When $r = 100$, it implies $n_{\text{eff}} = 9.91 \times 10 \approx 99$, which our data shows is insufficient for reliable estimates.

Figure 3 demonstrates the problem. Panel A shows success rates (fraction of configurations achieving $\rho \geq 0.95$) binned by effective sample size. Below $n_{\text{eff}} = 350$, success rates are erratic; above this threshold, they stabilise at 100%.

Panel B shows where the proportional-only model fails. At low reach values, the implied n_{eff} from $k \cdot \sqrt{r}$ produces success rates well below the 0.95 target.

We determine the floor more rigorously using logistic regression: fitting $P(\rho \geq 0.95) = \text{logistic}(\beta_0 + \beta_1 \cdot n_{\text{eff}})$ and inverting to find the n_{eff} where predicted success probability reaches 95%. This yields $n_{\min} = 359$ (95% CI: $[322, 408]$), confirming the binned analysis while providing uncertainty quantification (Table 1).

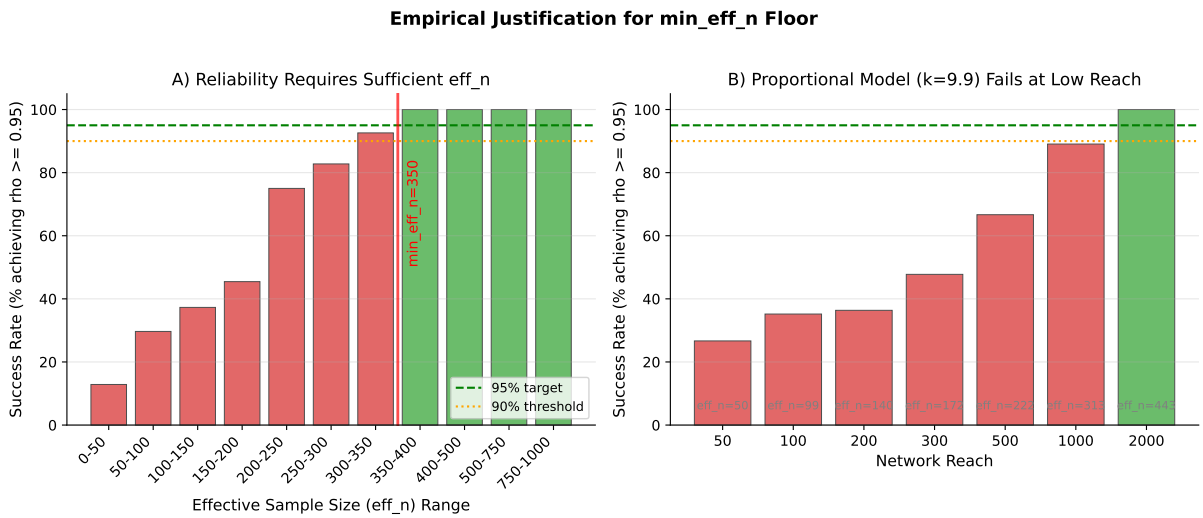

Figure 3: **Justifying the minimum floor.** (A) Success rate by n_{eff} bin. Red bars indicate bins below 0.95 success; the vertical line marks the fitted floor at 350. (B) The proportional model alone fails at low reach, justifying the floor constraint.

4.5 The Complete Model

Combining the proportional scaling with the minimum floor yields our final model:

$$n_{\text{eff}} = \max\left(9.91 \cdot \sqrt{r}, 350\right) \tag{7}$$

The sampling probability follows directly:

$$p = \min\left(1, \frac{n_{\text{eff}}}{r}\right) \tag{8}$$

Figure 4 shows the complete model. Panel A displays how the required n_{eff} transitions from the floor regime (flat at 350) to the proportional regime (scaling as $\sqrt{r}$) at the crossover point $r = 1247$. Panel B shows the corresponding sampling probability, which decreases as $1/\sqrt{r}$ once past the crossover.

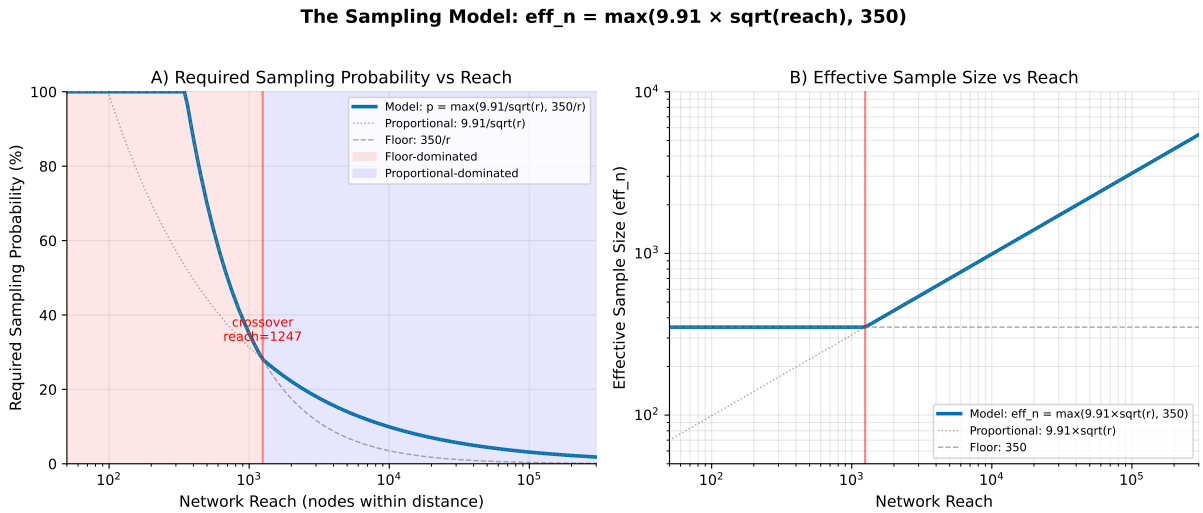

Figure 4: **The complete sampling model.** (A) Required effective sample size: floor regime (flat, $n_{\text{eff}} = 350$) transitions to proportional regime ($n_{\text{eff}} = 9.91 \cdot \sqrt{r}$) at reach ≈ 1247. (B) Required sampling probability: decreases as reach increases, enabling greater speedup for large-scale analysis.

Table 1 summarises the fitted model parameters.

Table 1: Fitted sampling model parameters.

Parameter	Value	Range	Description
k	9.91	[7.32, 13.86]	Proportional scaling constant
$n_{\min}$	350	[322, 408]	Minimum effective sample size
Crossover reach	1247	–	Where $k\sqrt{r} = n_{\min}$
$\hat{\alpha}$	0.5529	[0.4029, 0.6609]	Estimated power exponent

Model: $n_{\text{eff}} = \max(k \cdot \sqrt{r},\ n_{\min})$, where r is the network reach. Sampling probability: $p = \min(1,\ n_{\text{eff}}/r)$.
k: 75th percentile of implied values across configurations; range shows [mean, 95th percentile].
$n_{\min}$: logistic regression inversion at 95% success rate; range shows 95% CI.
$\hat{\alpha}$: estimated from general power model $n_{\text{eff}} = k \cdot r^{\alpha}$; not significantly different from 0.5 ($p = 0.38$), supporting the fixed $\sqrt{r}$ specification.

4.6 Model Selection

To justify the proposed $\sqrt{r}$ + floor specification against alternatives, we compare 5 candidate models: plain $\sqrt{r}$, $\sqrt{r}$ + floor (proposed), general power r^{α}, general power + floor, and logarithmic. Models are fitted by maximum likelihood using a logistic observation model and evaluated by AIC and leave-one-topology-out cross-validated Brier score (Table 2).

The General power model ($n_{\text{eff}} = k \cdot r^{\alpha}$, with $\hat{\alpha} = 0.5529$) achieves the lowest AIC (850.0), while the proposed $\sqrt{r}$ + floor model has $\Delta\text{AIC} = 11.6$. However, cross-validated Brier scores are

similar across all models (0.2123 vs. 0.2180), indicating that differences in out-of-sample prediction are modest—the AIC advantage does not translate into meaningfully better generalisation to held-out topologies. We retain the $\sqrt{r}$ specification for three reasons: (1) the hypothesis test does not reject $\alpha = 0.5$ ($p = 0.38$), so the data are consistent with square root scaling; (2) fixing $\alpha = 0.5$ provides a direct connection to sampling theory (SE $\propto 1/\sqrt{n}$), making the model interpretable rather than purely empirical; and (3) the floor component addresses a qualitatively distinct failure mode at low reach that no single power law can capture, even though the floor's contribution to in-sample fit is small. This is a modelling choice driven by interpretability and robustness rather than in-sample fit.

Table 2: Model comparison for effective sample size prediction. Models ranked by AIC; ΔAIC < 2 indicates substantial support.

Model	**Formula**	n_p	**AIC**	**ΔAIC**	**CV Brier**
General power	$k\,r^{\alpha}$	2	850.0	0.0	0.2123
Power + floor	$\max(k\,r^{\alpha}, m)$	3	853.2	3.2	0.2145
Square root	$k\sqrt{r}$	1	859.0	9.0	0.2195
Logarithmic	$k_1 + k_2 \log r$	2	861.0	11.0	0.2156
Sqrt + floor (proposed)	$\max(k\sqrt{r}, m)$	2	861.6	11.6	0.2180

n_p: number of fitted parameters. CV Brier: leave-one-topology-out cross-validated Brier score (lower is better). ΔAIC < 2: substantial support; 2–10: some support; > 10: essentially no support.

5 Adaptive Sampling Algorithm

5.1 Algorithm

Our adaptive sampling algorithm calibrates the sampling probability for each distance threshold based on the fitted model. Because reach varies with distance, optimal sampling probability also varies: short distances (low reach) may require full computation, while long distances enable aggressive sampling. Algorithm 1 presents the complete procedure.

Algorithm 1 Adaptive Per-Distance Sampling for Network Centrality

Require: Network $G = (V, E)$, distance thresholds $D = \{d_1, \ldots, d_m\}$
Require: Model parameters $k = 9.91$, $n_{\min} = 350$
Ensure: Centrality estimates $\hat{C}[d]$ for each $d \in D$

```
Phase 1: Reach Estimation
for each distance d ∈ D do
    Compute reach r_d ← mean nodes within distance d
end for
Phase 2: Adaptive Sampling
for each distance d ∈ D do
    Compute effective sample size: n_eff,d ← max(k · √r_d, n_min)
    Compute sampling probability: p_d ← min(1, n_eff,d / r_d)
    Sample sources S_d ⊆ V with probability p_d
    Compute centrality from S_d: Ĉ[d] ← CENTRALITY(G, S_d, d)
    Scale estimates: Ĉ[d] ← Ĉ[d] / p_d
end for
return {Ĉ[d] : d ∈ D}
```

The reach r_d is typically a byproduct of the centrality computation itself; when a quick estimate is needed beforehand, a pilot sample of ≈ 100 nodes distributed on a square kilometre

grid suffices. Sources are sampled uniformly at random, and sampled values are scaled by $1/p$ to produce unbiased estimates. This scaling is exact for betweenness (each source contributes independently) and approximate for harmonic closeness, though our floor constraint ($n_{\text{eff}} \geq 350$) ensures p is never low enough for the approximation to matter.

5.2 Computational Complexity

With sampling probability p, we compute from pN sources instead of N, reducing single-source computations by a factor of $1/p$. The theoretical speedup at high reach is $\sqrt{r}/k$, since $p = k/\sqrt{r}$—larger networks benefit more from sampling. For a network with distances spanning 500m to 20km, the model prescribes $p = 1.0$ at 500m (reach ≈ 200, no benefit), $p \approx 0.26$ at 5km (reach $\approx 3{,}000$), and $p \approx 0.07$ at 20km (reach $\approx 40{,}000$), yielding a theoretical 14× speedup at the longest—and most expensive—distance.

6 Validation

6.1 Experimental Protocol

We validate the model on the GLA and Greater Madrid networks described in Section 3. For each distance threshold, we compute ground-truth centrality ($p = 1.0$) and sampled centrality at probabilities from 1% to 100%, measuring Spearman ρ between the two. Each configuration is repeated with 3 random seeds; we report the mean ± standard deviation.

6.2 GLA Results

Figure 5 shows validation results for the Greater London network. At all three distances, the model-recommended probability achieves $\rho > 0.98$, exceeding the 0.95 target. Observed ρ consistently exceeds model predictions, demonstrating appropriate conservatism.

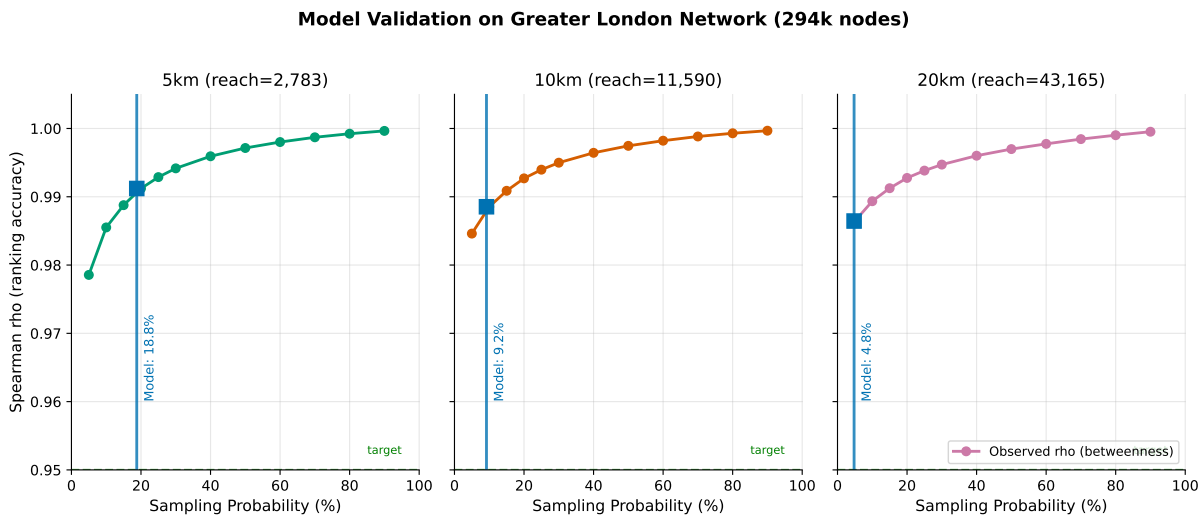

Figure 5: **GLA network validation.** (A) Accuracy vs. sampling probability at 5km, 10km, and 20km distances. Markers indicate the model-recommended probability; all exceed the 0.95 target. (B) Model conservatism: observed ρ exceeds predictions at all distances.

Table 3 presents detailed results. At 5km (reach = 2,783), observed $\rho = 0.991$ with 5.3× speedup. At 10km (reach = 11,590), $\rho = 0.989$ with 10.9× speedup. At 20km (reach = 43,165), $\rho = 0.986$ with 21.0× speedup. Standard deviations across replications are small (< 0.005 in ρ), indicating that a single sampled computation is sufficient for most purposes.

Table 3: Model Validation on Greater London Network

Distance	Reach	Model p	Observed ρ	Speedup	$\rho \geq 0.95$?
5km	2,783	18.8%	0.9912	5.3×	✓
10km	11,590	9.2%	0.9885	10.9×	✓
20km	43,165	4.8%	0.9864	21.0×	✓

Network: Greater London Area, 294,486 nodes, 20km live node buffer.

The validation demonstrates successful extrapolation well beyond synthetic training data: from ≈ 2,000-node training networks to 294,000 nodes, from maximum training reach of ≈ 2,600 to 43,165, and from controlled synthetic topologies to heterogeneous real-world structure. Despite these extrapolations, predictions remain conservative, suggesting that the n_{eff}-accuracy relationship is a robust property of network centrality rather than an artifact of specific configurations.

6.3 External Validation: Madrid

To test generalisability beyond London, we applied the identical protocol to the Greater Madrid metropolitan street network. The model achieves $\rho > 0.95$ for both closeness and betweenness at all tested distances (Figure 6), confirming that the fitted parameters generalise across morphologically distinct European cities.

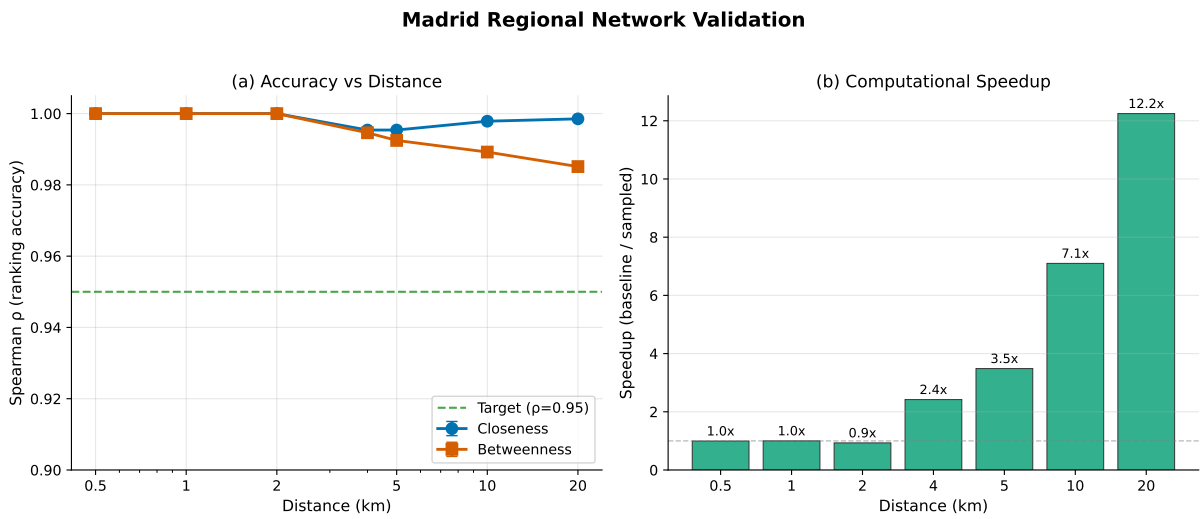

Figure 6: **Madrid external validation.** (A) Accuracy vs. distance for closeness and betweenness centrality. (B) Speedup at each distance.

7 Practical Guidelines

7.1 Applying the Model

For users of the `cityseer` library, the adaptive sampling functions implement the model automatically (see Appendix A.5). For other frameworks, the procedure is: (1) estimate reach r at each analysis distance, (2) compute $n_{\text{eff}} = \max(9.91 \cdot \sqrt{r}, 350)$ and $p = \min(1, n_{\text{eff}}/r)$, and (3) compute centrality from a random p-fraction of sources, scaling results by $1/p$.

Figure 7 provides visual lookup charts for common scenarios, and Table 4 gives a numerical reference for representative reach values.

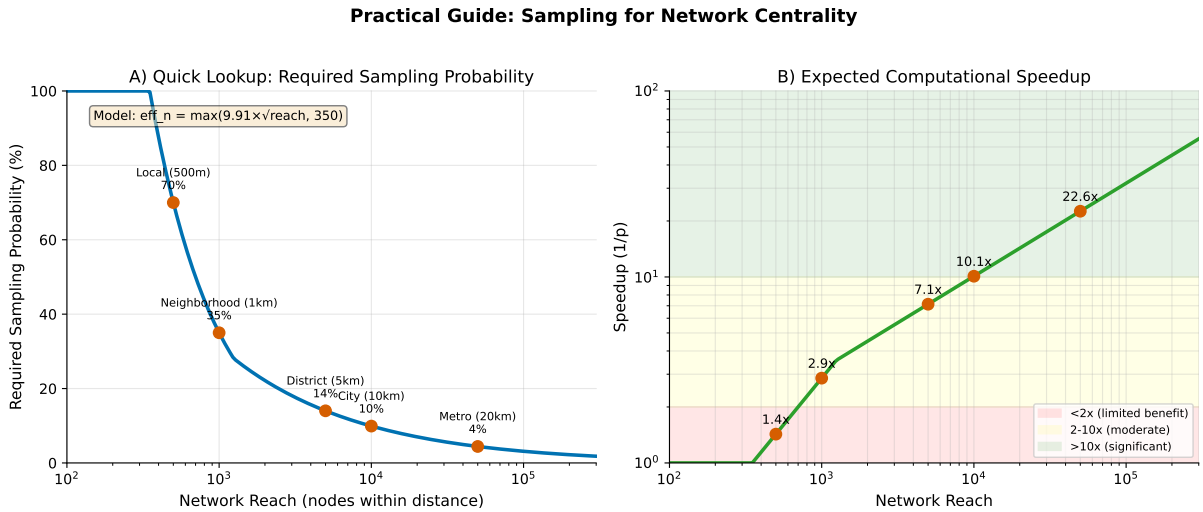

Figure 7: **Practical guidance.** (A) Required sampling probability by network reach, with common urban scenarios annotated. (B) Expected computational speedup.

Table 4: Practical Sampling Lookup Table

Reach	**eff_n**	p (%)	**Speedup**
100	350	100.0	1.0×
200	350	100.0	1.0×
500	350	70.0	1.4×
1,000	350	35.0	2.9×
2,000	443	22.2	4.5×
5,000	701	14.0	7.1×
10,000	991	9.9	10.1×
20,000	1401	7.0	14.3×
50,000	2216	4.4	22.6×
100,000	3134	3.1	31.9×

Usage: Find your network reach (nodes within analysis distance), read off the required sampling probability.
Model: $n_{\text{eff}} = \max(9.91 \cdot \sqrt{r}, 350)$, $p = n_{\text{eff}}/r$.

7.2 Rules of Thumb and Edge Cases

As a rough guide: at reach below 1247, use full computation—sampling overhead outweighs savings. At reach 1,000–10,000, sample 10–30% (3–10× speedup). At reach above 10,000, sample 5–15% (7–20× speedup). For applications requiring higher accuracy than $\rho \geq 0.95$, increase n_{eff} by a factor of 1.5–2× or average 2–3 replications with different random seeds.

Sampling is not appropriate when exact centrality values (not approximations) are required, when centrality is needed for specific nodes rather than the full network, or when the network is small enough (< 1,000 nodes) that full computation is already fast.

8 Discussion

8.1 Model Assumptions and Limitations

Our model rests on several assumptions. First, we assume uniform random sampling of source nodes. Non-uniform strategies—stratified by location or weighted toward high-centrality sources—could potentially achieve higher accuracy with fewer samples, but would require information about node importance that is unavailable before computation. Second, the model treats the network as statistically homogeneous: accuracy depends on global reach, not local structure. Real networks have heterogeneous topology, with dense urban cores surrounded by sparse suburbs. Our validation on the GLA and Madrid networks, which exhibit such heterogeneity, shows the model remains conservative overall, but accuracy may vary locally. For applications requiring spatially uniform accuracy, local calibration or stratified sampling could be explored. Third, we fitted the model primarily to betweenness centrality, which is more demanding than harmonic closeness. The model is therefore conservative for closeness, where lower sampling probabilities would suffice. Finally, our experiments used shortest-path distance; angular (simplest-path) distance, common in space syntax analysis, may exhibit different accuracy characteristics and warrants systematic validation.

8.2 Comparison with Theoretical Bounds

Theoretical work on centrality approximation (Riondato and Kornaropoulos, 2016) provides worst-case bounds on approximation error as a function of sample size. These bounds are typically conservative, requiring many more samples than our empirical analysis suggests are sufficient. The gap arises because theoretical bounds must hold for adversarial graph structures, while real urban networks have regular, well-behaved topology. Our approach trades worst-case guarantees for practical efficiency, achieving high accuracy with fewer samples by exploiting this regularity. For applications where worst-case bounds are required, theoretical approaches should be preferred; for typical urban analysis, our empirical model provides more practical guidance.

8.3 Practical Considerations

Sampling introduces computational overhead beyond the centrality computation itself: random source selection, reach estimation, and result scaling. For very short distances (low reach), this overhead may exceed the savings. Our floor constraint ($n_{\text{eff}} \geq 350$) mitigates this by preventing sampling when benefit is minimal, but analysts should be aware that the crossover point ($r \approx 1247$) represents the minimum reach where sampling provides meaningful speedup.

In multi-scale analysis, which computes centrality at many distance thresholds simultaneously, computational savings are uneven: short distances may dominate total runtime even when long distances are heavily sampled. Analysts should consider whether all distance thresholds are necessary, or whether interpolation from a subset of distances could provide adequate resolution.

8.4 Future Directions

Several extensions could improve this work. Adaptive per-node sampling, where the probability varies with local reach, could improve efficiency in heterogeneous networks. Importance sampling, weighting toward high-centrality sources, could reduce variance. Online calibration—adjusting sampling probability during computation based on observed variance—could provide tighter adaptation. Systematic validation for angular distance and extension to other centrality measures (eigenvector centrality, PageRank) would broaden applicability.

9 Conclusion

We have presented a two-parameter model that determines the sampling probability for network centrality approximation from network reach alone. Formal testing confirms the square root scaling assumption, and model comparison across five candidate functional forms supports the proposed specification. Validation on the Greater London (294,000 nodes) and Madrid street networks achieves Spearman $\rho > 0.98$ at all tested distances (5–20km), with speedups of 5.3–21.0×—demonstrating that the model extrapolates well beyond the synthetic networks on which it was trained.

The practical value lies in providing analysts with a calibrated way to trade computation time for approximation error, with confidence that sampled results preserve ranking accuracy. The largest benefits accrue at long distances on large networks—precisely where computational cost is highest. More broadly, our results suggest that the effective sample size–accuracy relationship is a consistent property of real-world network centrality rather than an artifact of specific configurations, though further validation on non-European cities would strengthen this claim.

The model is implemented in the open-source `cityseer` Python library.[1] Fitting scripts, validation data, and model comparison analyses are available in the repository's `analysis/sampling/` directory.

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A Technical Details

A.1 Synthetic Network Generation

Synthetic networks are generated using tile-based templates. Each template defines a connectivity pattern within a single tile; tiles are then arrayed to create networks of the desired extent.

Trellis A regular grid with diagonal connections, representing highly connected urban cores. Average degree ≈ 4.

Tree A hierarchical branching structure where each node connects to at most 3 others. Represents suburban cul-de-sac development. Average degree ≈ 2.3.

Linear A constrained grid with fewer cross-connections than trellis. Intermediate connectivity. Average degree ≈ 3.

Networks are tiled 24×24 to achieve approximately 12km extent, then the largest connected component is extracted. Node positions are perturbed slightly to avoid perfectly regular spacing.

A.2 GLA Network Preprocessing

The Greater London Authority road network is sourced from Ordnance Survey Open Roads data (GeoPackage format).

Preprocessing steps:

1. Remove invalid and empty geometries
2. Explode multi-part geometries
3. Build NetworkX graph from edges
4. Remove filler nodes (degree-2 nodes that don't represent junctions)

5. Remove dangling nodes (dead-end segments < 20m)
6. Consolidate nearby nodes (threshold 1m)
7. Extract largest connected component

Final network: ≈ 294,000 nodes.

A.3 Correctness Verification Against NetworkX

To ensure the correctness of our Rust-based centrality implementation in `cityseer`, we conducted systematic verification against the widely-used NetworkX library (Hagberg et al., 2008). For each synthetic network topology and distance threshold combination, we computed both harmonic closeness and betweenness centrality using our Rust implementation (via `cityseer` Python bindings) and NetworkX's reference functions.

For each node v in the test networks, we computed the absolute difference $\Delta_v = |C_{\text{cityseer}}(v) - C_{\text{NetworkX}}(v)|$. The maximum difference across all nodes and all test configurations was $\max_v \Delta_v < 10^{-6}$, attributable to floating-point arithmetic precision.

Verification was performed across all three synthetic topologies, distance thresholds from 200m to 3200m, both shortest-path and angular distance heuristics, and both harmonic closeness and betweenness centrality.

A.4 Hájek Estimator

When estimating population totals from samples, two classical approaches exist: the Horvitz-Thompson (H-T) estimator and the Hájek estimator. We selected the Hájek estimator based on theoretical considerations.

The Horvitz-Thompson estimator (Horvitz and Thompson, 1952) estimates a population total as $\hat{T}_{HT} = \sum_{i \in S} y_i/\pi_i$, where S is the sample and π_i is the inclusion probability. The Hájek estimator (Hájek, 1971) instead uses a ratio form:

$$\hat{T}_H = N \cdot \frac{\sum_{i \in S} y_i/\pi_i}{\sum_{i \in S} 1/\pi_i} \tag{9}$$

where N is the known population size. For simple random sampling with equal inclusion probabilities, the Hájek estimator has variance $\text{Var}(\hat{T}_H) \approx \text{Var}(\hat{T}_{HT}) \cdot (1 - n/N)$. Since the population size (total number of network nodes) is known, the Hájek estimator is the natural choice. In our implementation, centrality contributions are accumulated unscaled during traversal, then multiplied by N/n (total nodes divided by sampled nodes) after all sources have been processed.

A.5 Implementation in `cityseer`

The sampling model is implemented in the `cityseer` Python library (version 4.23+). The adaptive function automatically probes reachability, computes sampling probabilities from the fitted model, and scales results:

```
from cityseer.metrics import networks

gdf = networks.node_centrality_shortest_adaptive(
    network_structure,
    nodes_gdf,
    distances=[500, 1000, 2000, 5000, 10000],
    compute_closeness=True,
    compute_betweenness=True,
    target_rho=0.95
)
```

For manual control, a fixed sampling probability can be specified via the `sample_probability` parameter of `node_centrality_shortest`. Helper functions for custom workflows (`compute_required_p`, `probe_reachability`, `spatial_sample`) are available in the `cityseer.config` module; see the library documentation for details.

A.6 Power Exponent Diagnostics

Figure 8 presents bootstrap diagnostics for the power exponent α in the general model $n_{\text{eff}} = k \cdot r^{\alpha}$. The estimated $\hat{\alpha} = 0.5529$ is consistent with the theoretical value of 0.5.

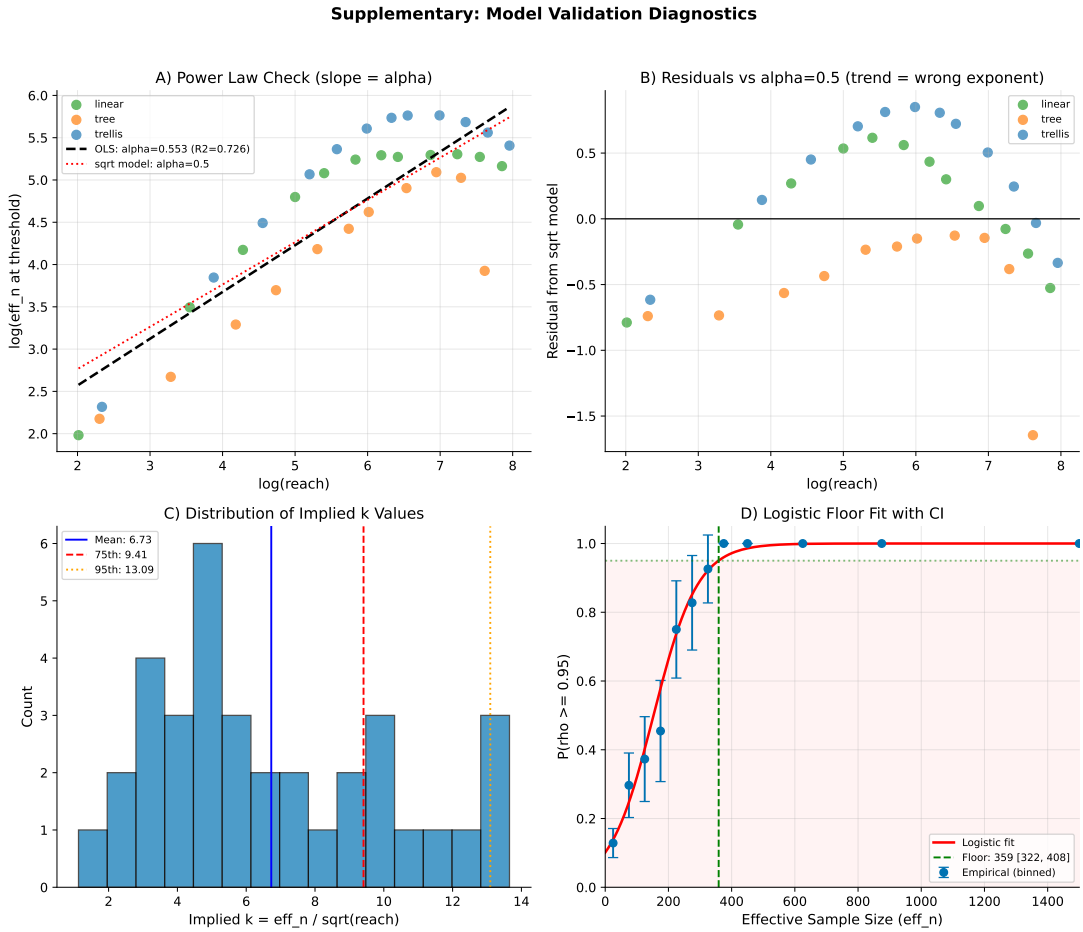

Figure 8: **Power exponent diagnostics.** Bootstrap distribution of the estimated exponent $\hat{\alpha}$. The theoretical value $\alpha = 0.5$ falls within the 95% confidence interval.

A.7 Reproducibility

All analysis scripts, fitting data, and validation results are available at `https://github.com/benchmark-urbani` Model parameters are stored in `output/sampling_model.json` and synced to `pysrc/cityseer/config.py` for library integration.